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Smart Football: A Data-Driven Approach to Fair Match Organisation for Communities

6CM995 Project Plan

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2 Introduction

Football is the world's most widely played sport; in England it is deeply implanted in English community life. However, in England its roots in community are slowly growing further away. Not only in rising ticket prices but also digitally. Digital innovation in local football management has not kept pace with the professional game. Local organisers still rely on group chats, spreadsheets, and informal arrangements that often lead to scheduling conflicts and uneven team formation.

Beyond those challenges, amateur football teams are also more susceptible to bias in player selection. Team organisers may subconsciously favour friends or frequent participants, resulting in unbalanced matches and lower satisfaction among players.

In face of these issues, this project proposes Smart Football, a data-driven scheduling platform designed to reduce bias and improve fairness in community football. By automating player allocation and match organisation through transparent, data-based processes, the system aims to prove how technology can support inclusion, efficiency, and enjoyment at the amateur level.

3 Aim And Objectives

3.1 Aim

To design and evaluate a data-driven web platform that supports fair and efficient community football organisation through automated team formation, player self-assessment, and scheduling tools.

3.2 Objectives

1. Develop a secure login and player registration system.
2. Implement match scheduling and attendance tracking functionality.
3. Develop a fairness algorithm that uses player information and feedback to create balanced teams.
4. Create a dashboard for visualising match and attendance statistics.
5. Evaluate the system for functionality, usability, and fairness.

4 Methodology And Approach

The web application will be developed using Python (Django) for the backend and HTML, CSS, and JavaScript for the frontend, with SQLite used for data storage during development. The fairness algorithm will be implemented using scikit-learn and Pandas, and Chart.js will provide data visualisation for match and attendance statistics.

The platform will be designed as a responsive web application to ensure accessibility across desktop and mobile browsers. The feature set was defined following consultation with a local football organiser, who highlighted ease of use, fairness in team formation, and accessibility for players with varying experience levels.

The project will follow an Agile development approach, dividing the work into short iterative sprints. Each sprint will focus on implementing a key feature such as registration, scheduling, or team balancing. This approach is ideal as it allows for frequent testing, rapid feedback on the prototype, and flexibility to refine the fairness algorithm as the system evolves.

Evaluation will involve three levels of testing: functional testing to verify usability and correctness; algorithm fairness checks to measure team balance; and performance testing to assess response times.

Supervisor meetings will be used to review sprint progress and ensure the project remains aligned with its objectives

4.1 System Requirements

- **Platform Type:** Web-based platform accessible using desktop and mobile browsers.
- **Match Format:** 5-a-side games allowing players to register and join sessions.
- **Player Profiles:** Each player can self-assess skill, position preference, experience, and fitness level.
- **Ranking System:** A community driven rating system will adjust player levels based on peer feedback and match participation.
- **Team Balancing:** The platform will use data from player profiles and peer ratings to generate fair teams automatically.
- **Match Management:** Organisers can create sessions, manage attendance, and record performance ratings.
- **Payment Context:** Matches typically involve a small participation fee (£5), but finances is outside the project scope.

5 Ethical Considerations

No personal or sensitive data will be collected. All datasets will be artificially generated and removed after project submission. Data handling will follow University of Derby ethical standards.

6 Timeline and Gantt Chart

Key milestones are detailed in the 'Expected Outcome' column of the timeline.

Week (2025– 2026)	Focus Area	Planned Activities	Expected Outcome
Wk 6–7 (Oct– Nov)	Planning & Ethics	Finalise project idea, meet supervisor, submit ethics form, background reading.	Approved ethics form and refined project scope.
Wk 8 (Nov 7)	Project Plan Submission	Write and submit project plan.	Approved plan and timeline confirmed.
Wk 9–11 (Nov– Dec)	Core Setup	Install Django, set up database, create login and registration system.	Functional prototype with working authentication.
Wk 12–14 (Dec– Jan)	Scheduling Module	Implement match creation, attendance tracking, and organiser dashboard.	Scheduling and attendance fully functional.
Wk 15 (Jan 12– 16)	Prototype Review	Show progress to supervisor, collect feedback.	Feedback received, direction confirmed.
Wk 16–20 (Jan – Mar)	Fairness Algorithm & Data Integration	Develop team-balancing algorithm, test data-driven ranking, add visualisation using Chart.js.	Balanced team allocation feature implemented.
Wk 21–22 (Mar)	Poster Preparation	Create and submit poster summarising design and findings.	Poster and presentation ready.
Wk 23–25 (Apr)	Testing & Refinement	Conduct functional, usability, and fairness testing; bug fixes.	Polished final system and test results.
Wk 26–27 (Apr– May)	Documentation	Write final dissertation chapters, reflect on results.	Complete dissertation.
Wk 28 (May 7)	Submission	Submit dissertation	

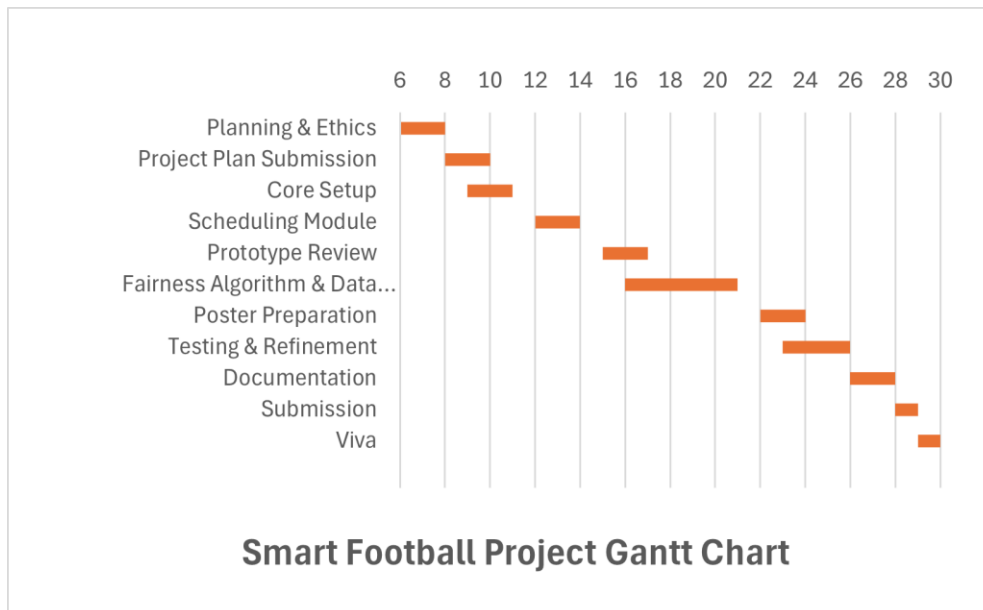


Figure 1. Gantt chart showing weekly project timeline (Weeks 6–29).

7 Risks

Risk	Impact	Solution
Integration complexity	Medium	Build and test modules separately before integration
Algorithm bias or inaccuracy	High	Use varied synthetic data
Time overruns	High	Weekly sprints and supervisor check-ins
Data loss	Low	Version control and regular backups

8 Initial Reading List

Bunker, R. and Thabtah, F. (2019) 'A machine learning framework for sport result prediction', *Applied Computing and Informatics*, 15(1), pp. 27-33.

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